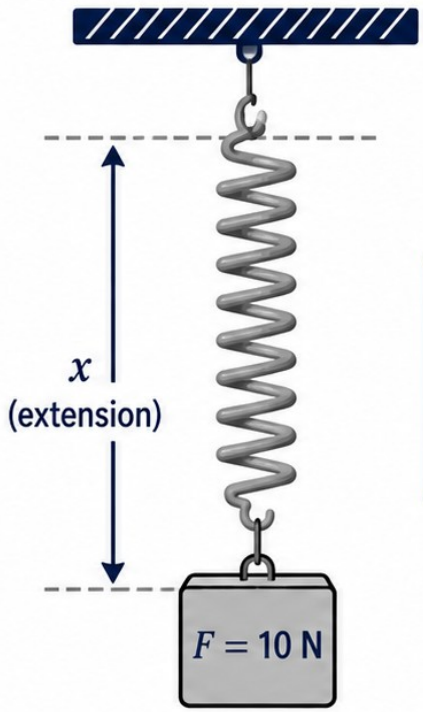
Sensitivity and Uncertainty Propagation



- How uncertain is a prediction when one input is not known exactly?
- Input range → model → output range
- Use physical reasoning before MATLAB

MODEL

$$x = \frac{F}{k}$$

INPUT RANGE

$k = 95\text{–}105$ N/m

→

MODEL

$$x = \frac{F}{k}$$

→

OUTPUT RANGE

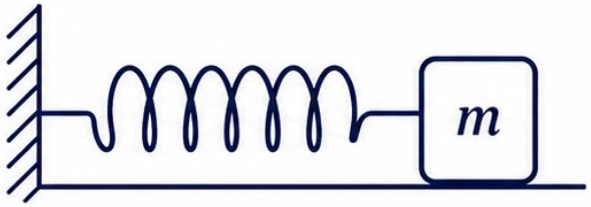
predicted extension x

Hold $F = 10$ N fixed; vary one plausible input.

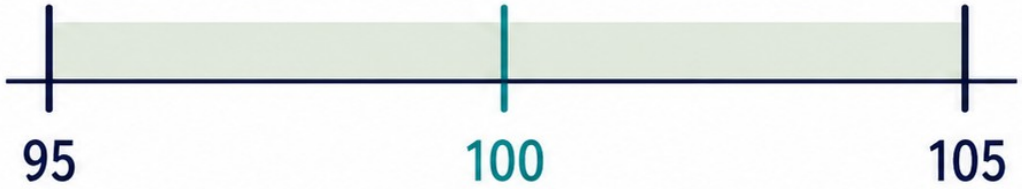
One Familiar Model, a New Question

Ideal Hooke's law: $x = F/k$

Fixed load: $F = 10$ N



Spring stiffness, k (N/m)

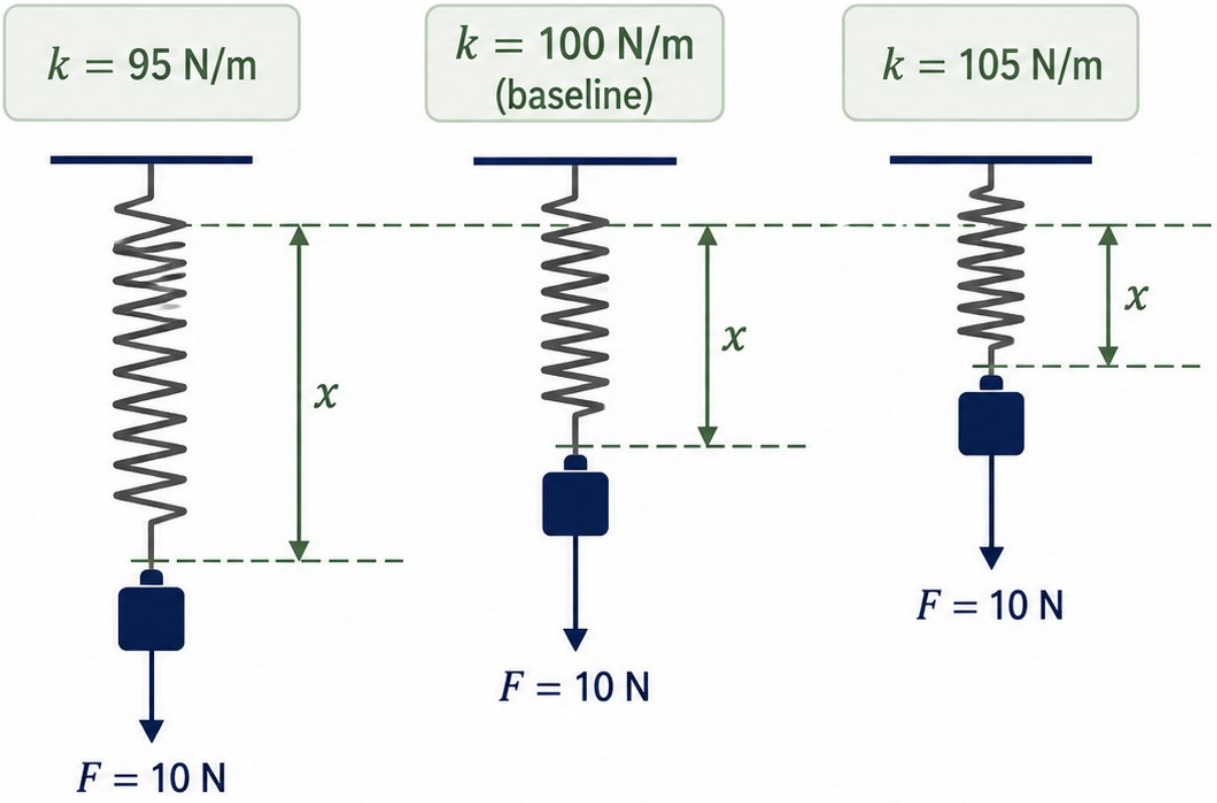


$k = 95–105$ N/m

plausible range around 100 N/m

Week 4 varied a parameter; here the range represents input uncertainty.

Predict Before Computing



Lower k should give larger extension

Higher k should give smaller extension

The output range should straddle 0.10 m

$$x = \frac{F}{k}$$

F fixed at 10 N

Baseline, Lower, and Upper Values

CASE	SPRING STIFFNESS k	WHAT CHANGES?
LOWER	95 N/m	k only
BASELINE	100 N/m	reference case
UPPER	105 N/m	k only

Use lower, baseline, and upper input values

Hold force, equation, units and assumptions fixed

HOLD FIXED



Force $F = 10$ N

Equation $x = F/k$

Units

Assumptions

CHANGE ONE THING



Spring stiffness k

THIS IS ONE-AT-A-TIME COMPARISON

One-at-a-Time Algorithm

1 Set baseline and plausible range



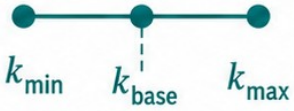
F 🔒
k ——●——●——
x 🔒

2 Hold all other inputs fixed



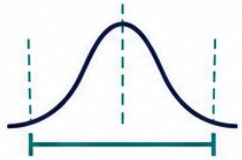
F 🔒
k ——●——●——
x 🔒

3 Evaluate three cases



F 🔒
k
x 🔒

4 Report output range and validate baseline



F 🔒
k
x 🔒

5 Interpret without false precision



F 🔒
k
x 🔒

One varying input is the core discipline.

Trace the MATLAB Scaffold

```
stiffness_cases_Npm = [95 100 105]

extension_cases_m = force_baseline_N./stiffness_cases_Npm

min and max report the range
```

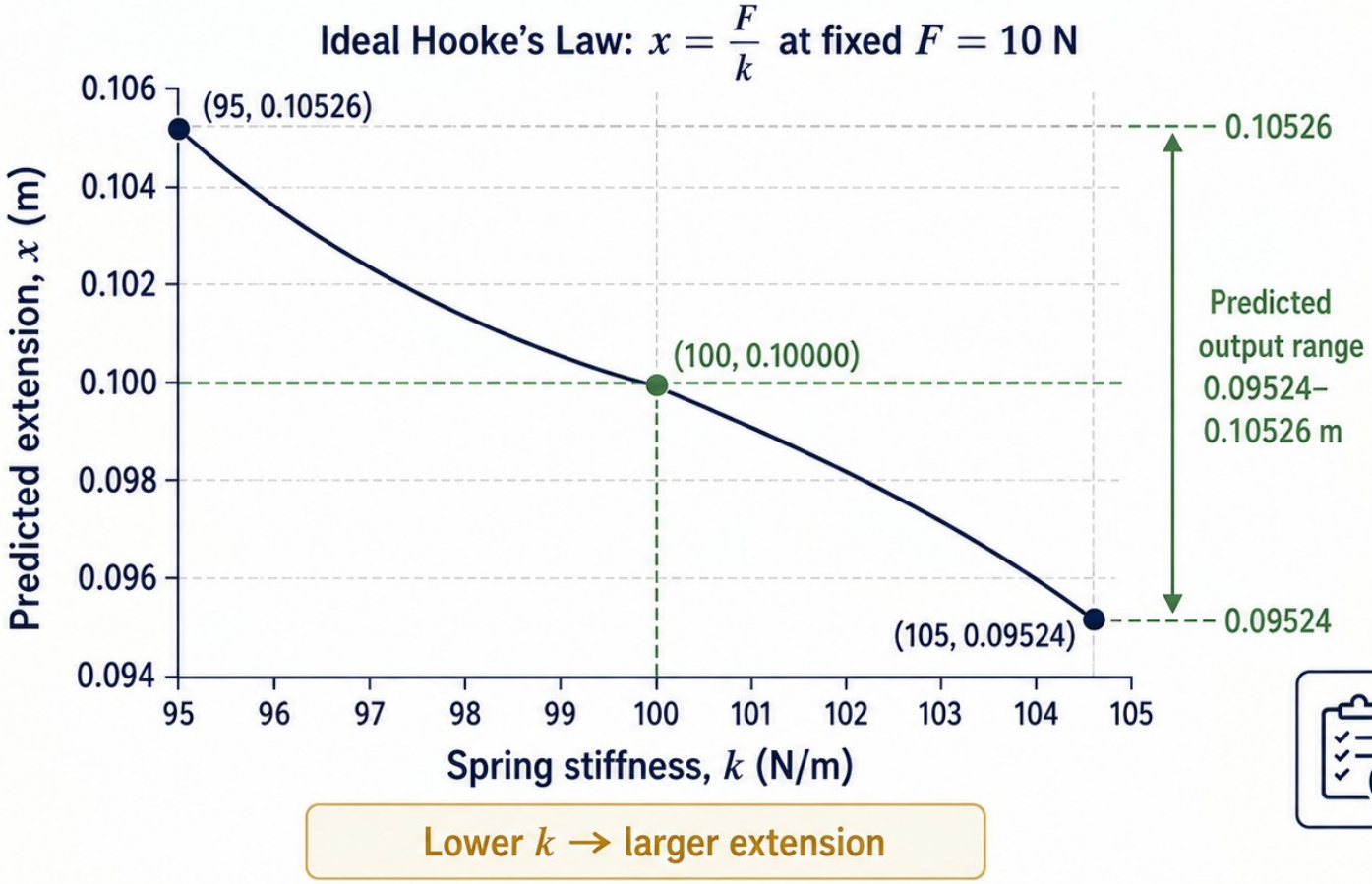
`./` element-wise division

$$95 \longrightarrow x(1)$$

$$100 \longrightarrow x(2)$$

$$105 \longrightarrow x(3)$$

What Range Does the Model Predict?



Baseline:
0.10000 m

Input range:
$k = 95–105$ N/m

Predicted range:
0.09524–0.10526 m

Controlled parameter cases, not random trials

Report the Result Without False Precision

WEAK REPORT

$$x = 0.10000 \text{ m}$$

The value alone hides the input range.

DEFENSIBLE REPORT

For $F = 10$ N and $k = 95–105$ N/m, the ideal model predicts $x \approx 0.095–0.105$ m.



- 0.10000 m alone hides the input range
- State the load, stiffness range and output range
- **Extra decimal places do not reduce uncertainty**

Core Validation: Check the Baseline



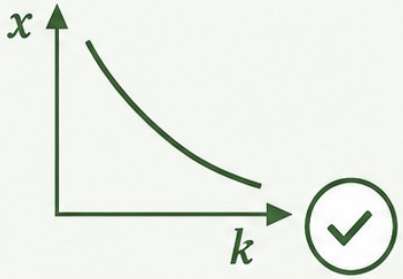
Hand calculation

$10/100 = 0.10$ m



Unit check

$N/(N/m) = m$



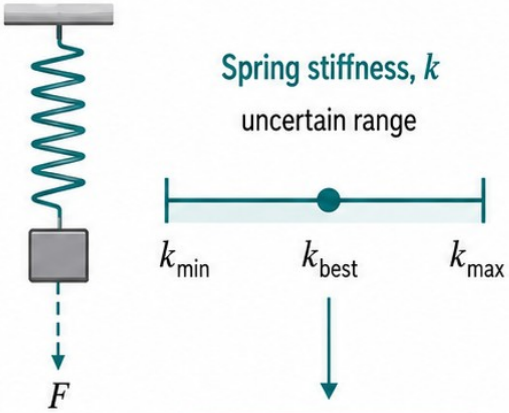
Trend check

Lower k gives higher x

A smooth graph alone is not validation.

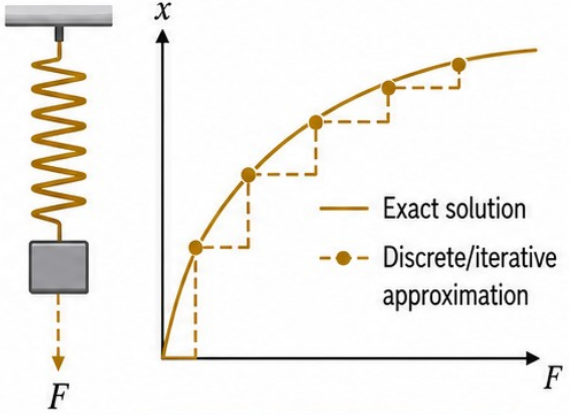
Three Different Reasons a Prediction Can Be Imperfect

1 Parameter uncertainty: uncertain input value



Different plausible k values lead to different predicted extensions.

2 Numerical approximation error: finite timestep, grid or iteration

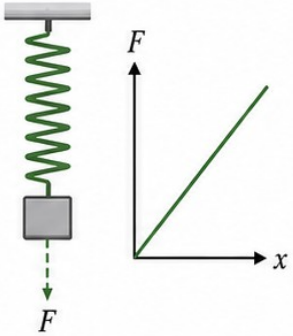


The discrete solution approaches the exact solution, but some approximation error remains.

This direct Hooke calculation has no meaningful timestep, grid or iteration approximation error.

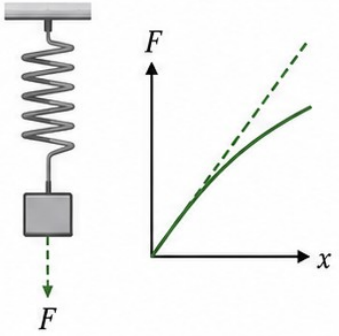
3 Model limitation: assumptions do not fully represent reality

Ideal spring model



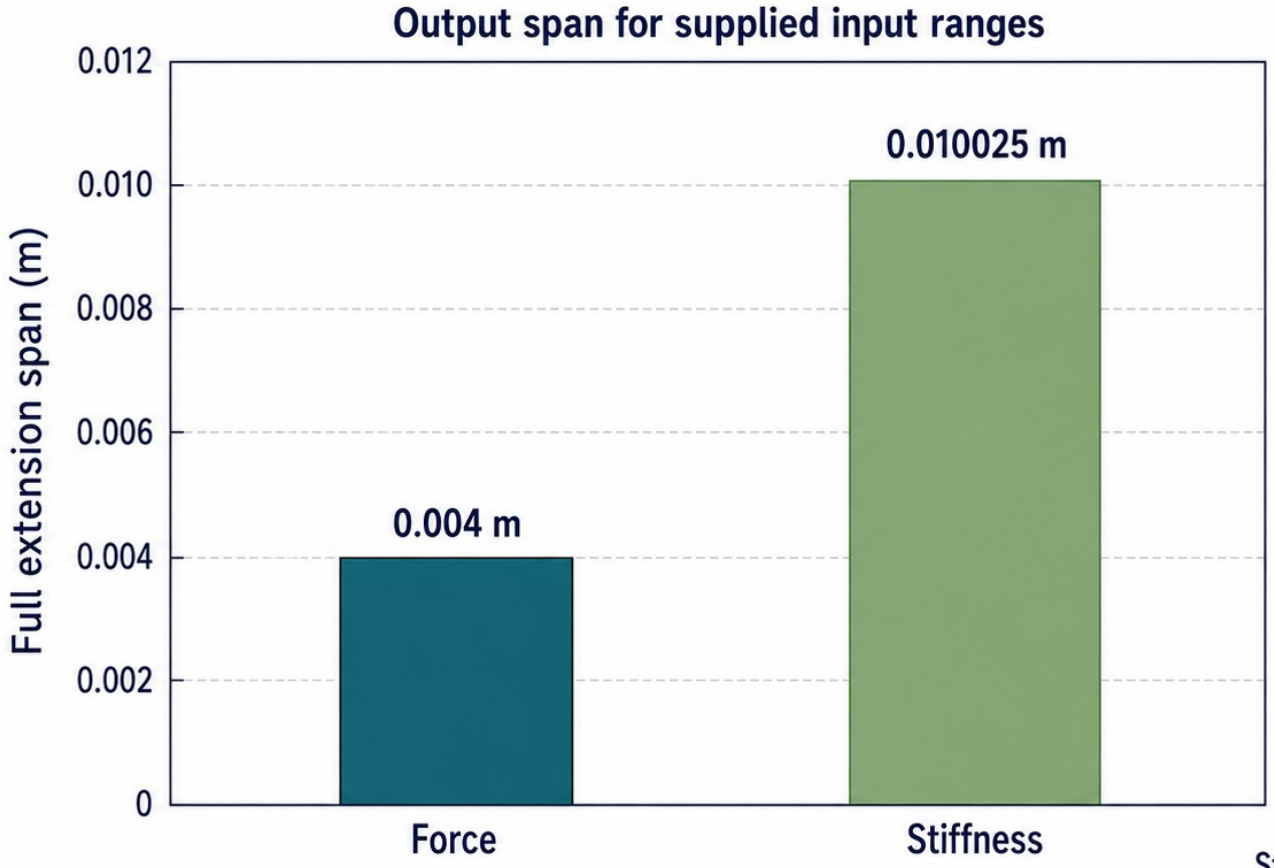
Assumes linear, elastic, no damping, small deformation, etc.

Real (imperfect) spring



Nonlinearity, hysteresis, damping, preload, or other effects present.

Working Exposure: Compare a Second Uncertain Input



FORCE SWEEP

Hold $k = 100$ N/m
Vary $F = 9.8$–10.2 N

Force output range:
0.098–0.102 m

Hold $k = 100$ N/m;
vary $F = 9.8$–10.2 N

Force output range:
0.098–0.102 m

Stiffness gives the larger effect for these supplied ranges

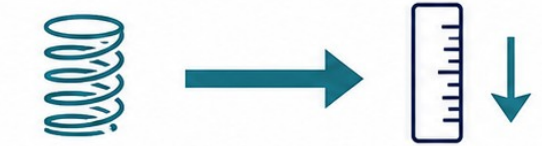
Student modification: predict, then calculate.

Working Exposure: Percentage Sensitivity

$$S \approx (\Delta x/x_0)/(\Delta p/p_0)$$

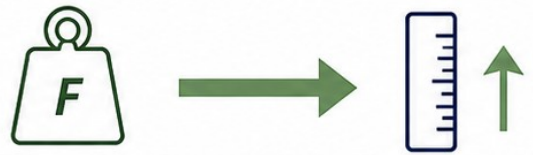
Stiffness sensitivity

$$S_k \approx -1$$



Force sensitivity

$$S_F = +1$$



Sign gives direction; magnitude gives fractional responsiveness

Working exposure: Interpret supplied sensitivities; do not derive.

Working Exposure: Supplied First-Order Propagation

$$\frac{u_x}{x} \approx \sqrt{\left(\frac{u_F}{F}\right)^2 + \left(\frac{u_k}{k}\right)^2}$$

- $\frac{u_x}{x}$: Relative uncertainty in extension, x
- $\left(\frac{u_F}{F}\right)^2$: Relative uncertainty in force, F
- $\left(\frac{u_k}{k}\right)^2$: Relative uncertainty in spring stiffness, k

$u_F = 0.2$ N; $u_k = 5$ N/m

Combined estimate $\approx 5.39\%$, $u_x \approx 0.00539$ m

Assumes small independent uncertainties.

Exit Ticket: What Can You Defend?



1 · INPUT RANGE

Name one uncertain input and plausible range



2 · HOLD FIXED

State what stays fixed



3 · DEFENSIBLE CONCLUSION

Give output range and classify a limitation

Write one concise statement you can defend.

parameter uncertainty • numerical approximation error • model limitation